

Activity 1

NAME: _____**Q.1.** Simplify the expression $5x - [3 - 4(x - 2)]$ **Q.2.** Simplify the expression $\frac{3x}{2} \div [10y \div \frac{5y}{6x}]$

Q.3. Convert to scientific notation: -0.00052

Q.4. Simplify the expression $\left(\frac{3x^2y}{z}\right)^2 \cdot \left(\frac{y^{-1}z}{x^5}\right)$ and write your answer with only positive exponents.

Q.5. Simplify the expression $\sqrt[3]{16 \cdot x^5 \cdot y^3} \cdot \sqrt[3]{x \cdot y}$.

Q.6. Rationalize the denominator $\frac{2x^2y}{\sqrt[3]{y^4}}$.

Q.7. Factor the polynomial $x^6 - 27y^3$.

Q.8. Expand and simplify the product of polynomials $(3x - 4y) \cdot (xy - 6x^2)$.

Q.9. Express the following algebraic expression as a polynomial:

$$(\sqrt{x} + \sqrt{y}) \cdot (\sqrt{x} - \sqrt{y})$$